

# Детекция объектов

Занятие 18

Computer Vision

Лектор  
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**r\_d**

# OBJECT DETECTION

Что такое **детекция объектов**?

- **Локализация**
  - Где находится объект?
- **Классификация**
  - Что это за объект?

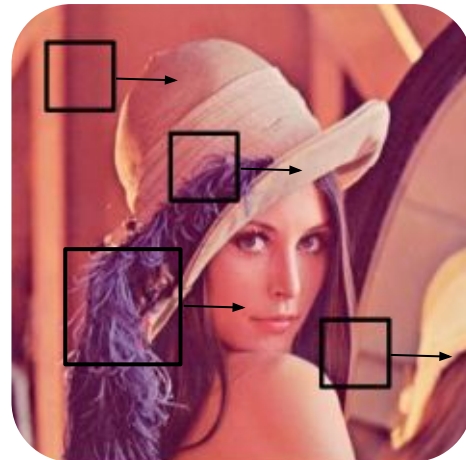


[source1](#) [source2](#)

# OBJECT DETECTION

Помните детектор Виолы-Джонса?

- Sliding window approach
- Для нескольких масштабов
- **Проблема**
  - Находит только один класс объектов



# BOUNDING BOX

bbox  $\rightarrow$  [x, y, width, height]

- Tightest box surrounding object
- In 2D or 3D

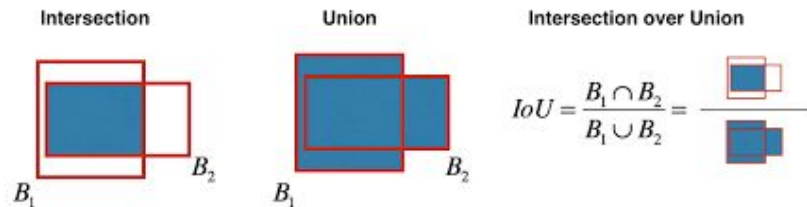


<https://algolux.com/resources/snow-fall/>

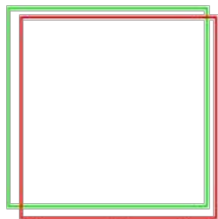
# IOU METRIC (INTERSECTION OVER UNION)

How do we measure detection accuracy?

- IoU (a.k.a. Jaccard index)

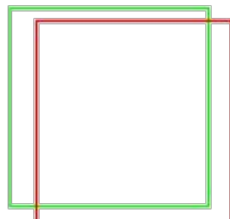


IoU=0.92



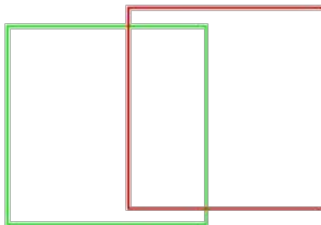
Excellent

IoU=0.71



Good

IoU=0.39



Poor

Detection vs Target

<https://www.interstellarengine.com/ai/Intersection-Over-Union.html>

<https://www.oreilly.com/library/view/hands-on-convolutional-neural/9781789130331/a0267a8a-bd4a-452a-9e5a-8b276d7787a0.xhtml>

# R-CNN

- Regions with CNN features (paper)

## R-CNN: *Regions with CNN features*

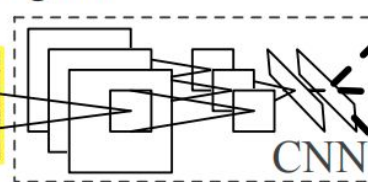


1. Input image

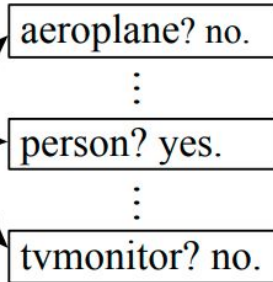


2. Extract region proposals (~2k)

warped region



3. Compute CNN features

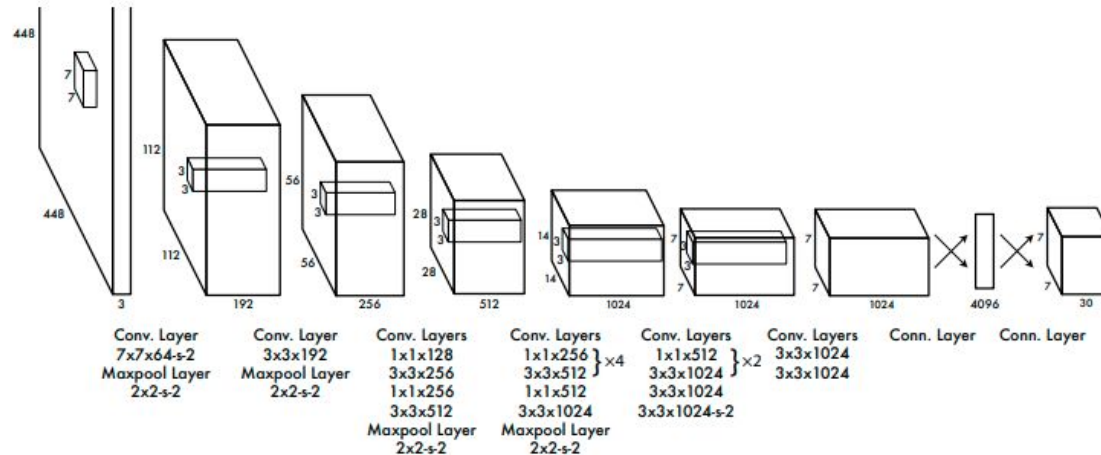


4. Classify regions

# SSD & YOLO

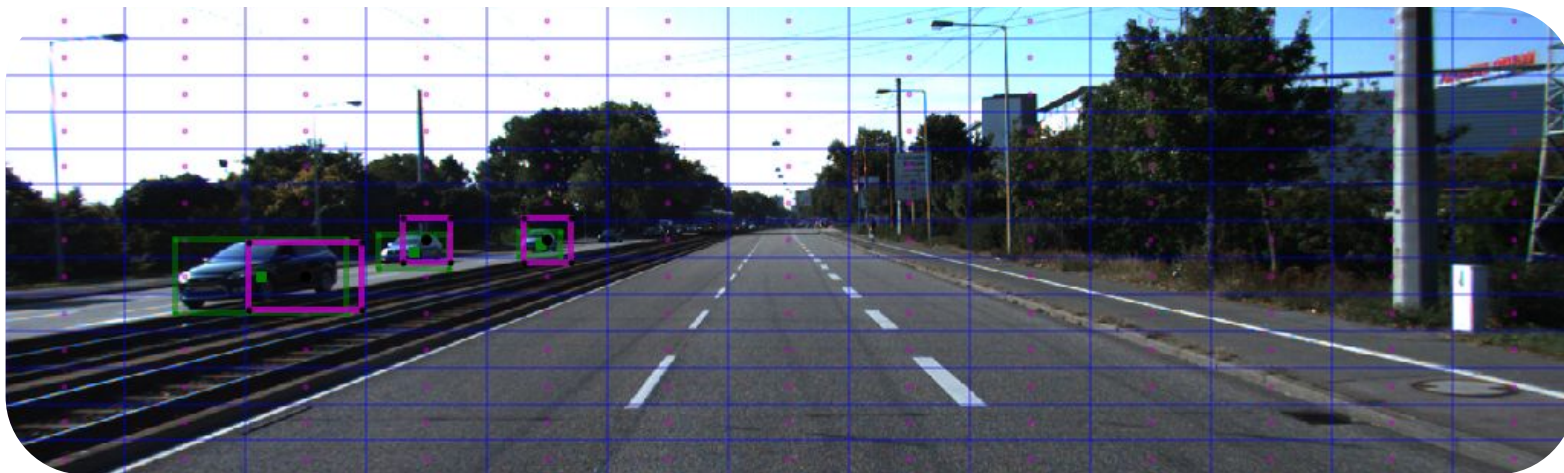
Вычисляют все изображение **сразу**, не требуют никаких region proposals

- Single Shot Detector (SSD)
- You Only Look Once (YOLO)



# MULTI-BOX HEAD

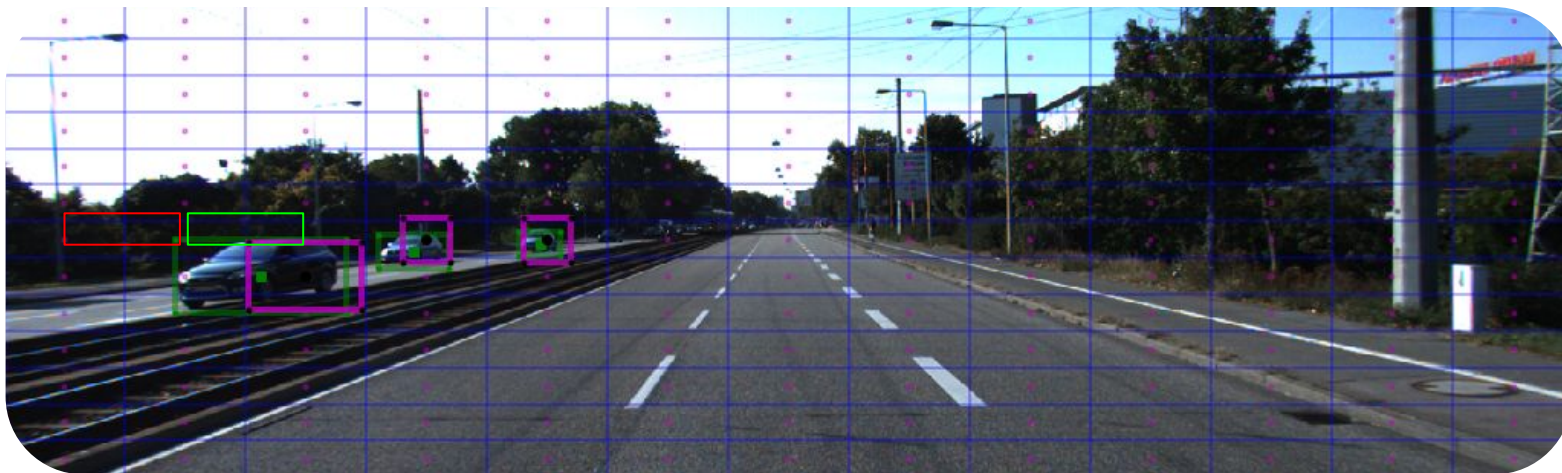
- Anchor boxes (anchors)





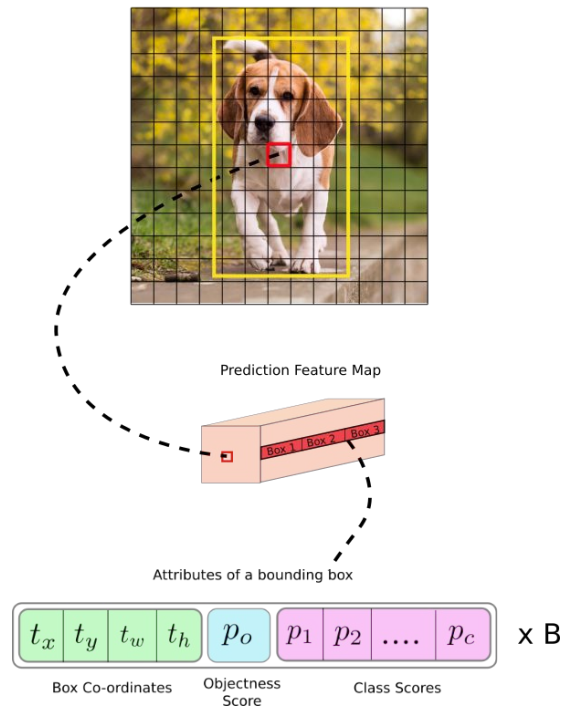
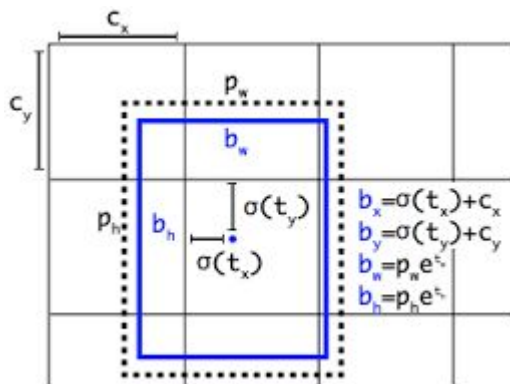
# MULTI-BOX HEAD

- Anchor boxes (anchors)

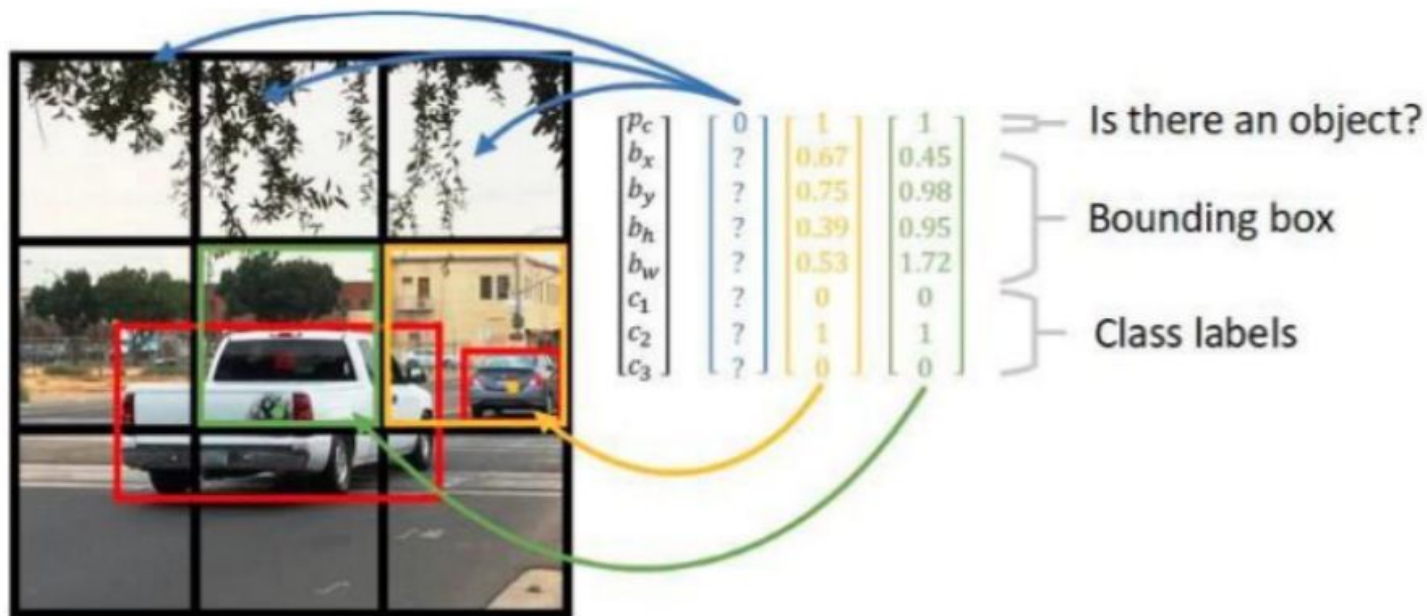


# MULTI-BOX HEAD

- For each anchor box the detector predicts
  - Offset with respect to the anchor center
  - Box inflation/deflation rates
  - Anchor “objectness”
  - Classification probabilities



# MULTI-BOX HEAD



# YOLO LOSS FUNCTION

During the training the following loss is minimized:

$$\begin{aligned}
 & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[ (x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
 & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[ \left( \sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left( \sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\
 & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\
 & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\
 & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2
 \end{aligned}$$

1 when there is object, 0 when there is no object

Bounding Box Location (x, y) when there is object

Bounding Box size (w, h) when there is object

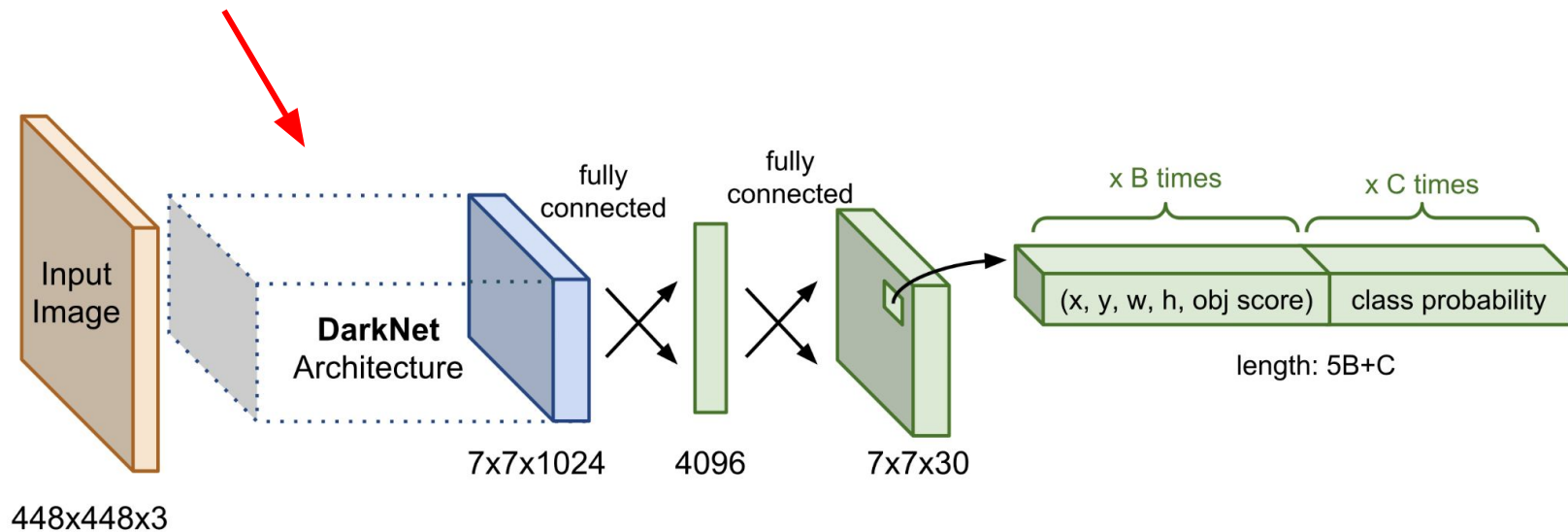
Confidence when there is object

1 when there is no object, 0 when there is object

Confidence when there is no object

Class probabilities when there is object

# BACKBONE



Q&A

???



ЗАВЖДИ Є КУДИ  
ЗРОСТАТИ